Akmal Ashirmatov

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Research interests: LLM post-training, reasoning, test-time scaling, learning from environment feedback, efficient inference

Education

- **ETH Zürich** *M.Sc. Computer Science, GPA 5.93/6.0* *Sep. 2024 – May 2027 (exp.)*
  - MSc thesis (in progress) in the Learning & Adaptive Systems (LAS) group of Prof. Andreas Krause.
  - Selected coursework: Probabilistic AI, Advanced ML, Reliable and Trustworthy AI, Large-Scale AI Engineering.
- **Technical University of Munich** *M.Sc. Data Engineering & Analytics (1 semester)* *Apr. 2024 – Sep. 2024*
- **Lomonosov MSU Tashkent** *B.Sc. Applied Mathematics & CS, with Honors, GPA 5.00/5.00* *2018 – 2022*
  - Thesis: Increasing the efficiency of person recognition in images with artificial neural networks.

Publications

* equal contribution

[P1] **On Repulsive and Attractive Teachers: Separating Correctness from Behavior in Self-Distillation** ↗
Anton Baumann, **Akmal Ashirmatov**, Leo Schmidt-Traub, Frederike Lübeck, Jonas Hübotter, Thomas Kleine Buening, Andreas Krause
FLLMPT Workshop at NeurIPS 2026

[P2] **Test-Time Scaling with Weak Verifiers via Self-Play** ↗
Akira Yoshiyama*, **Akmal Ashirmatov***, Jonas Hübotter, Ido Hakimi, Andreas Krause
DL4C Workshop at ICML 2026

[P3] **ESCAPE: Explaining Stable Diffusion via Cross-Attention Maps and Prompt Editing** ↗
Diego Zafferani*, Giovanni De Muri*, Johanna Hedlund Lindmar*, **Akmal Ashirmatov***, Sinie van der Ben, Rita Sevastjanova, Mennatallah El-Assady
VISxAI Workshop at IEEE VIS 2025

Research Experience

- **MSc Thesis, LAS, ETH Zürich** *with F. Lübeck, A. Baumann, A. Krause* *Oct. 2026 – May 2027 (exp.)*
  - Post-training LLMs to reason about their uncertainty.
- **Research Assistant, LAS, ETH Zürich** *with J. Hübotter, F. Lübeck, T. Kleine Buening, A. Krause* *Mar. 2026 – Jul. 2026*
  - Characterized behavioral biases in LLM self-distillation; showed contrastive self-distillation mitigates them and improves reasoning without an auxiliary GRPO objective [P1].
- **Semester Project, LAS, ETH Zürich** *with J. Hübotter, I. Hakimi, A. Krause* *Dec. 2025 – May. 2026*
  - Co-trained a proposer and a solver via self-play for code generation from weak verifiers and environment feedback; beats context evolvement and rejection sampling at test time [P2].
- **Course Research Project, IVIA Lab, ETH Zürich** *with M. El-Assady* *Mar. 2025 – Jul. 2025*
  - Built a framework to study how geometry-preserving prompt edits change cross-attention in Stable Diffusion [P3].

Industry Experience

- **Google, London** *SWE Intern, Applied Research team* *Jun. 2026 – Sep. 2026*
  - Designed and prototyped a generative tool for Gemini Live and contributed production code to the Gemini app.
  - Wrote the design document for integrating the tool into the existing agent harness.
- **Huawei, Zurich Research Center** *Research Intern* *Jun. 2025 – Sep. 2025*
  - LLM inference optimization with Triton and vLLM; profiling, ML compiler optimization, IR debugging.

Selected Projects

- **Distributed Fine-Tuning of Apertus for Legal Text** ↗ *Large-Scale AI Engineering, ETH* *2026*
  - LoRA/full fine-tuning of Apertus 8B/70B on legal data; multi-node DeepSpeed ZeRO-2/3 training.

Teaching & Service

- **Teaching Assistant, Probabilistic Artificial Intelligence** *ETH Zürich, A. Krause* *Sep. 2025 – Feb. 2026*
  - Built the course's new interactive demo on diffusion models.
- **Competitive Programming Coach and Organizer** *Lomonosov MSU Tashkent* *2019 – 2023*
  - Lectured on algorithms and trained ICPC teams; led Uzbekistan's team at APIO; chaired the Lomonosov Festival jury.

Honors & Awards

- **ICPC**: 3× national champion of Uzbekistan; 2× NERC 2nd-degree diploma (4th and 5th in Central Asia) *2019 – 2023*
- **Huawei iRunner**: 6× prize-winner in international heuristic optimization contests *2022 – 2025*
- Ranked in the top 1.3% of all participants in Meta Hacker Cup, top 1.9% in Google Code Jam, top 0.8% in CodeChef SnackDown; Codeforces Master ↗ *2021 – 2022*
- 1st place, Republican Olympiad in Informatics, Uzbekistan *2022*

Skills

- **ML/LLM:** *PyTorch, Hugging Face, vLLM, verl, LoRA/PEFT, W&B*
- **Systems:** *Python, C/C++, Triton, Bash, Linux, Docker, Kubernetes, Ray, DeepSpeed, GCP, HPC/Slurm*